POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2042 | Semester II | Theory | 4 Credits

Electronic Devices and Applications

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics, communication, biomedical, microelectronics and IC-design programme group listed in the official syllabus**
Contact hours	**60**
Teaching scheme	**4:0:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**6 h/week; 90 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2042**, titled **Electronic Devices and Applications**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

Source treatment: PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

Verified source note: No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

This course builds an operational understanding of AC–DC conversion, waveform shaping and transistor devices. It moves from diode circuits and filters to BJT, JFET, MOSFET and UJT characteristics used in electronic systems.

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Prerequisite foundation

Passive components, semiconductor theory and diodes (1041); Engineering Mathematics I (1002); Mathematics for Electrical Sciences (2001B).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
4	0	0	0	6 h/week; 90 h total	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Apply AC–DC conversion and linear waveform shaping.
2. Apply nonlinear waveform shaping and voltage multiplication.
3. Analyse BJT operation and characteristics.
4. Understand FET and UJT operation and characteristics.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Apply AC–DC conversion and linear waveform-shaping techniques in electronic systems.	Apply
CO2	Apply waveform shaping and voltage multiplication methods for signal conditioning.	Apply
CO3	Explain operation, configurations and characteristics of bipolar junction transistors.	Understand
CO4	Explain operation and characteristic behaviour of field-effect and unijunction transistors.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Linear Wave-Shaping Circuits	Half-wave, centre-tapped and bridge rectifiers; comparison; capacitor/inductor/pi filters; RC differentiator and integrator.	15 hours	CO1	Module 1
2	Nonlinear Wave-Shaping Circuits	Series, parallel, combinational and Zener clippers; biased/unbiased clampers; half/full-wave doubler and tripler.	15 hours	CO2	Module 2
3	Bipolar Junction Transistors	NPN/PNP structure; NPN operation; CB/CE/CC configurations; CB/CE characteristics; operating modes; load line and bias.	15 hours	CO3	Module 3
4	JFET, MOSFET and UJT	N-channel JFET; depletion MOSFET; drain and transfer characteristics; UJT structure and V–I behaviour.	15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Linear Wave-Shaping Circuits

Half-wave, centre-tapped and bridge rectifiers; comparison; capacitor/inductor/pi filters; RC differentiator and integrator.

CO1 · 15 hours

Nonlinear Wave-Shaping Circuits

Series, parallel, combinational and Zener clippers; biased/unbiased clampers; half/full-wave doubler and tripler.

CO2 · 15 hours

Bipolar Junction Transistors

NPN/PNP structure; NPN operation; CB/CE/CC configurations; CB/CE characteristics; operating modes; load line and bias.

CO3 · 15 hours

JFET, MOSFET and UJT

N-channel JFET; depletion MOSFET; drain and transfer characteristics; UJT structure and V–I behaviour.

CO4 · 15 hours

MODULE 1

Linear Wave-Shaping Circuits

Half-wave, centre-tapped and bridge rectifiers; comparison; capacitor/inductor/pi filters; RC differentiator and integrator.

15 hours **CO1** **Understand · Apply**

Learning goals

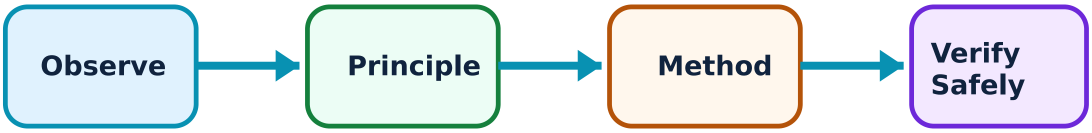
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Linear Wave-Shaping Circuits: identify the system, state its principle, follow the correct method and verify the result safely.

1. Half-wave, centre-tapped and bridge rectifiers−

A rectifier uses diode directionality to produce unidirectional output from AC input. Half-wave circuits use one half cycle; centre-tapped and bridge arrangements provide full-wave action through different circuit structures. Official comparison uses DC output voltage, ripple factor, peak inverse voltage and transformer utilisation factor.

Study and application method

Mark input polarity and diode state for each half cycle. Sketch output over one complete cycle before comparing circuits.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Mark input polarity and diode state for each half cycle. Sketch output over one complete cycle before comparing circuits.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Full-wave rectification is not perfectly smooth DC; unfiltered output still has ripple.

2. Filters in rectifier outputs−

Capacitor, inductance and pi filters reduce ripple by storing/releasing energy or opposing current variation in appropriate circuit positions. Their application is considered with rectifier type and load.

Study and application method

Trace capacitor charging/discharging in a simple filter. State that filtering improves smoothness but must respect component ratings and load behaviour.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Many electrolytic capacitors are polarity-sensitive; incorrect polarity can be hazardous.

3. RC differentiator and integrator−

An RC network can emphasise rapid input changes or accumulate input behaviour depending on time constant relative to signal period. The official focus is obtaining differentiated and integrated output waveforms.

Study and application method

Compare RC time constant with waveform duration, draw input/output on aligned time axes and explain capacitor charging.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare RC time constant with waveform duration, draw input/output on aligned time axes and explain capacitor charging.

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Common mistake / precaution: Labelled waveform axes and polarity are essential; a sketch without time reference is incomplete.

RC time-constant helper

Resistance (Ω)

1000

Capacitance (µF)

10

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Nonlinear Wave-Shaping Circuits

Series, parallel, combinational and Zener clippers; biased/unbiased clampers; half/full-wave doubler and tripler.

15 hours CO2 Understand · Apply

Learning goals

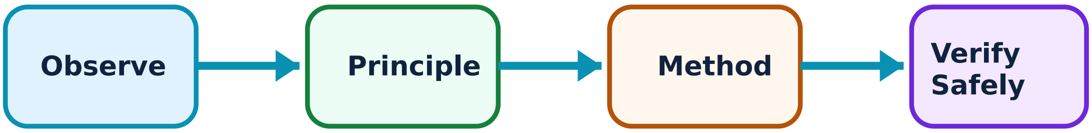
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Nonlinear Wave-Shaping Circuits: identify the system, state its principle, follow the correct method and verify the result safely.

1. Clipper circuits−

Clippers limit selected waveform portions using diode conduction and non-conduction. Series, parallel, combinational and Zener circuits differ by placement, bias and clipping level.

Study and application method

Determine diode condition for positive and negative portions separately. Include bias source or Zener threshold before drawing output.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Determine diode condition for positive and negative portions separately. Include bias source or Zener threshold before drawing output.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A clipper removes or limits amplitude; it does not necessarily shift the whole waveform baseline.

2. Clamper circuits−

Clampers shift a waveform reference level while ideally preserving shape. Positive and negative clampers are studied with and without bias. Capacitor charging and diode action establish clamping level.

Study and application method

Show assumed initial capacitor condition, identify fast charging interval and explain slower holding interval relative to time constant.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Show assumed initial capacitor condition, identify fast charging interval and explain slower holding interval relative to time constant.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse clamping with clipping: clamper action is mainly level shifting.

3. Voltage multipliers−

Diode-capacitor multiplier circuits build output levels from peak input voltage. The syllabus includes half-wave/full-wave doublers and tripler circuits.

Study and application method

Label capacitor polarities and trace charge transfer half-cycle by half-cycle. Mention load sensitivity in practical output interpretation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Label capacitor polarities and trace charge transfer half-cycle by half-cycle. Mention load sensitivity in practical output interpretation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Multiplier circuits can create voltages above a single supply peak; verify capacitor and diode ratings.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Bipolar Junction Transistors

NPN/PNP structure; NPN operation; CB/CE/CC configurations; CB/CE characteristics; operating modes; load line and bias.

15 hours **CO3** **Understand · Apply**

Learning goals

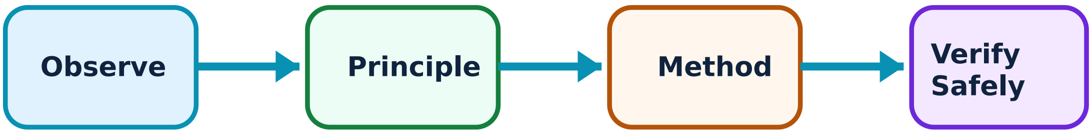
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bipolar Junction Transistors: identify the system, state its principle, follow the correct method and verify the result safely.

1. BJT structure and NPN operation−

A bipolar junction transistor has three regions and two junctions. NPN and PNP devices differ in carrier arrangement and supply polarity. Controlled base-emitter action permits collector-current control in an appropriate operating region.

Study and application method

Draw symbol and label emitter, base and collector. Explain current paths using a stated conventional-current direction and bias condition.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw symbol and label emitter, base and collector. Explain current paths using a stated conventional-current direction and bias condition.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not mix NPN and PNP polarities; symbol arrow conveys an important convention.

2. Configurations, modes and characteristics−

Common-base, common-emitter and common-collector configurations use a different terminal as reference. Input/output characteristic plots relate current and voltage. Cutoff, active and saturation are operating modes with meanings linked to junction bias and intended use.

Study and application method

Name axes before interpreting a graph. State whether operation is amplification or switching and connect that intent to active/cutoff/saturation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Name axes before interpreting a graph. State whether operation is amplification or switching and connect that intent to active/cutoff/saturation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A characteristic curve is not one fixed operating point; external circuit conditions select the point.

3. Load line and biasing−

The DC load line represents collector-current/voltage combinations imposed by external circuit. The Q point is chosen quiescent operation. Fixed and voltage-divider bias establish suitable DC conditions.

Study and application method

Draw load line from supply and load resistance, show Q point qualitatively and explain why bias stabilises operation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Bias is deliberate DC condition setting, not merely applying any voltage.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

JFET, MOSFET and UJT

N-channel JFET; depletion MOSFET; drain and transfer characteristics; UJT structure and V–I behaviour.

15 hours · CO4 · Understand · Apply

Learning goals

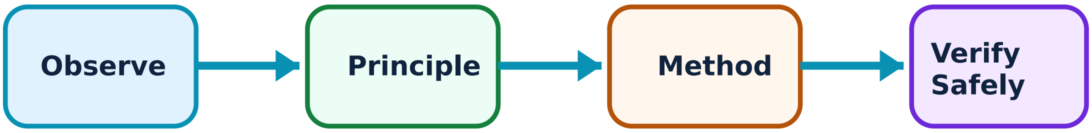
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for JFET, MOSFET and UJT: identify the system, state its principle, follow the correct method and verify the result safely.

1. FET and N-channel JFET−

A field-effect transistor controls channel current with an electric field. In an N-channel JFET, reverse gate bias changes depletion width and drain current.

Study and application method

Label gate, drain and source, distinguish control voltage from drain supply and read drain characteristics by regions/parameter curves.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Label gate, drain and source, distinguish control voltage from drain supply and read drain characteristics by regions/parameter curves.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High input impedance does not mean gate conditions and device ratings can be ignored.

2. N-channel depletion MOSFET−

A depletion MOSFET has an insulated gate structure and is described through drain and transfer characteristics. The syllabus focuses on N-channel depletion operation.

Study and application method

Use a terminal-labelled diagram and state qualitatively how gate control affects channel conduction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a terminal-labelled diagram and state qualitatively how gate control affects channel conduction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Insulated gate does not mean immunity to static damage; follow ESD precautions.

3. Unijunction transistor−

A unijunction transistor has a single PN junction and a negative-resistance region in its V–I characteristic. The official scope requires its structure and characteristic behaviour.

Study and application method

Draw bases and emitter, mark important V–I points and explain conduction change step by step.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw bases and emitter, mark important V–I points and explain conduction change step by step.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Negative-resistance region describes characteristic slope; it does not imply unlimited energy generation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Linear Wave-Shaping Circuits

Use the concept flow to revise observation, principle, method and verification.

Module 2: Nonlinear Wave-Shaping Circuits

Use the concept flow to revise observation, principle, method and verification.

Module 3: Bipolar Junction Transistors

Use the concept flow to revise observation, principle, method and verification.

Module 4: JFET, MOSFET and UJT

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare rectifier/DC-equipment charts and AC-to-DC conversion flowcharts.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Simulate and compare RC differentiator and integrator waveforms.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Tabulate biased/unbiased clipper circuits and draw clamper/voltage-multiplier diagrams.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Collect BJT/FET/UJT data sheets and prepare symbol, terminal and parameter charts.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Linear Wave-Shaping Circuits

Explain Half-wave, centre-tapped and bridge rectifiers and state one correct method, application or precaution.−

A rectifier uses diode directionality to produce unidirectional output from AC input. Half-wave circuits use one half cycle; centre-tapped and bridge arrangements provide full-wave action through different circuit structures. Official comparison uses DC output voltage, ripple factor, peak inverse voltage and transformer utilisation factor.

Answer framework: Mark input polarity and diode state for each half cycle. Sketch output over one complete cycle before comparing circuits.

Important point: Full-wave rectification is not perfectly smooth DC; unfiltered output still has ripple.

Explain Filters in rectifier outputs and state one correct method, application or precaution.−

Capacitor, inductance and pi filters reduce ripple by storing/releasing energy or opposing current variation in appropriate circuit positions. Their application is considered with rectifier type and load.

Answer framework: Trace capacitor charging/discharging in a simple filter. State that filtering improves smoothness but must respect component ratings and load behaviour.

Important point: Many electrolytic capacitors are polarity-sensitive; incorrect polarity can be hazardous.

Explain RC differentiator and integrator and state one correct method, application or precaution.−

An RC network can emphasise rapid input changes or accumulate input behaviour depending on time constant relative to signal period. The official focus is obtaining differentiated and integrated output waveforms.

Answer framework: Compare RC time constant with waveform duration, draw input/output on aligned time axes and explain capacitor charging.

Important point: Labelled waveform axes and polarity are essential; a sketch without time reference is incomplete.

Module 2 — Nonlinear Wave-Shaping Circuits

Explain Clipper circuits and state one correct method, application or precaution.−

Clippers limit selected waveform portions using diode conduction and non-conduction. Series, parallel, combinational and Zener circuits differ by placement, bias and clipping level.

Answer framework: Determine diode condition for positive and negative portions separately. Include bias source or Zener threshold before drawing output.

Important point: A clipper removes or limits amplitude; it does not necessarily shift the whole waveform baseline.

Explain Clamper circuits and state one correct method, application or precaution.−

Clampers shift a waveform reference level while ideally preserving shape. Positive and negative clampers are studied with and without bias. Capacitor charging and diode action establish clamping level.

Answer framework: Show assumed initial capacitor condition, identify fast charging interval and explain slower holding interval relative to time constant.

Important point: Do not confuse clamping with clipping: clamper action is mainly level shifting.

Explain Voltage multipliers and state one correct method, application or precaution.−

Diode-capacitor multiplier circuits build output levels from peak input voltage. The syllabus includes half-wave/full-wave doublers and tripler circuits.

Answer framework: Label capacitor polarities and trace charge transfer half-cycle by half-cycle. Mention load sensitivity in practical output interpretation.

Important point: Multiplier circuits can create voltages above a single supply peak; verify capacitor and diode ratings.

Module 3 — Bipolar Junction Transistors

Explain BJT structure and NPN operation and state one correct method, application or precaution.−

A bipolar junction transistor has three regions and two junctions. NPN and PNP devices differ in carrier arrangement and supply polarity. Controlled base-emitter action permits collector-current control in an appropriate operating region.

Answer framework: Draw symbol and label emitter, base and collector. Explain current paths using a stated conventional-current direction and bias condition.

Important point: Do not mix NPN and PNP polarities; symbol arrow conveys an important convention.

Explain Configurations, modes and characteristics and state one correct method, application or precaution.−

Common-base, common-emitter and common-collector configurations use a different terminal as reference. Input/output characteristic plots relate current and voltage. Cutoff, active and saturation are operating modes with meanings linked to junction bias and intended use.

Answer framework: Name axes before interpreting a graph. State whether operation is amplification or switching and connect that intent to active/cutoff/saturation.

Important point: A characteristic curve is not one fixed operating point; external circuit conditions select the point.

Explain Load line and biasing and state one correct method, application or precaution.−

The DC load line represents collector-current/voltage combinations imposed by external circuit. The Q point is chosen quiescent operation. Fixed and voltage-divider bias establish suitable DC conditions.

Answer framework: Draw load line from supply and load resistance, show Q point qualitatively and explain why bias stabilises operation.

Important point: Bias is deliberate DC condition setting, not merely applying any voltage.

Module 4 — JFET, MOSFET and UJT

Explain FET and N-channel JFET and state one correct method, application or precaution.−

A field-effect transistor controls channel current with an electric field. In an N-channel JFET, reverse gate bias changes depletion width and drain current.

Answer framework: Label gate, drain and source, distinguish control voltage from drain supply and read drain characteristics by regions/parameter curves.

Important point: High input impedance does not mean gate conditions and device ratings can be ignored.

Explain N-channel depletion MOSFET and state one correct method, application or precaution.−

A depletion MOSFET has an insulated gate structure and is described through drain and transfer characteristics. The syllabus focuses on N-channel depletion operation.

Answer framework: Use a terminal-labelled diagram and state qualitatively how gate control affects channel conduction.

Important point: Insulated gate does not mean immunity to static damage; follow ESD precautions.

Explain Unijunction transistor and state one correct method, application or precaution.−

A unijunction transistor has a single PN junction and a negative-resistance region in its V–I characteristic. The official scope requires its structure and characteristic behaviour.

Answer framework: Draw bases and emitter, mark important V–I points and explain conduction change step by step.

Important point: Negative-resistance region describes characteristic slope; it does not imply unlimited energy generation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Linear Wave-Shaping Circuits.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Nonlinear Wave-Shaping Circuits.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Bipolar Junction Transistors.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: JFET, MOSFET and UJT.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Half-wave, centre-tapped and bridge rectifiers; comparison; capacitor/inductor/pi filters; RC differentiator and integrator..
2. Develop a complete answer or practical plan for: Series, parallel, combinational and Zener clippers; biased/unbiased clampers; half/full-wave doubler and tripler..
3. Develop a complete answer or practical plan for: NPN/PNP structure; NPN operation; CB/CE/CC configurations; CB/CE characteristics; operating modes; load line and bias..
4. Develop a complete answer or practical plan for: N-channel JFET; depletion MOSFET; drain and transfer characteristics; UJT structure and V–I behaviour..

LAST-MILE REVISION

Quick Revision

Module 1: Linear Wave-Shaping Circuits

Half-wave, centre-tapped and bridge rectifiers; comparison; capacitor/inductor/pi filters; RC differentiator and integrator.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Nonlinear Wave-Shaping Circuits

Series, parallel, combinational and Zener clippers; biased/unbiased clampers; half/full-wave doubler and tripler.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Bipolar Junction Transistors

NPN/PNP structure; NPN operation; CB/CE/CC configurations; CB/CE characteristics; operating modes; load line and bias.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: JFET, MOSFET and UJT

N-channel JFET; depletion MOSFET; drain and transfer characteristics; UJT structure and V–I behaviour.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Half-wave, centre-tapped and bridge rectifiers	A rectifier uses diode directionality to produce unidirectional output from AC input.
Filters in rectifier outputs	Capacitor, inductance and pi filters reduce ripple by storing/releasing energy or opposing current variation in appropriate circuit positions.
RC differentiator and integrator	An RC network can emphasise rapid input changes or accumulate input behaviour depending on time constant relative to signal period.
Clipper circuits	Clippers limit selected waveform portions using diode conduction and non-conduction.
Clamper circuits	Clampers shift a waveform reference level while ideally preserving shape.
Voltage multipliers	Diode-capacitor multiplier circuits build output levels from peak input voltage.
BJT structure and NPN operation	A bipolar junction transistor has three regions and two junctions.
Configurations, modes and characteristics	Common-base, common-emitter and common-collector configurations use a different terminal as reference.
Load line and biasing	The DC load line represents collector-current/voltage combinations imposed by external circuit.
FET and N-channel JFET	A field-effect transistor controls channel current with an electric field.
N-channel depletion MOSFET	A depletion MOSFET has an insulated gate structure and is described through drain and transfer characteristics.
Unijunction transistor	A unijunction transistor has a single PN junction and a negative-resistance region in its V–I characteristic.

LEARNING RESOURCES

References

Officially listed print resources

1. R. S. Sedha, A Text Book of Applied Electronics.
2. Boylestad, Electronic Devices and Circuits.
3. N. N. Bhargava, Kulshreshtha and S. C. Gupta, Basic Electronics and Linear Circuits.

Officially listed web resources

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